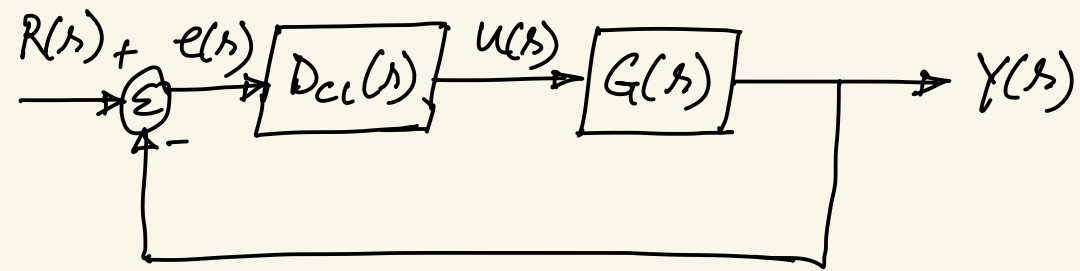




Notes on designing a PID Controller



PID controller is defined in terms of the error signal and the control input $U(s)$. The transfer function of the PID controller is as follows

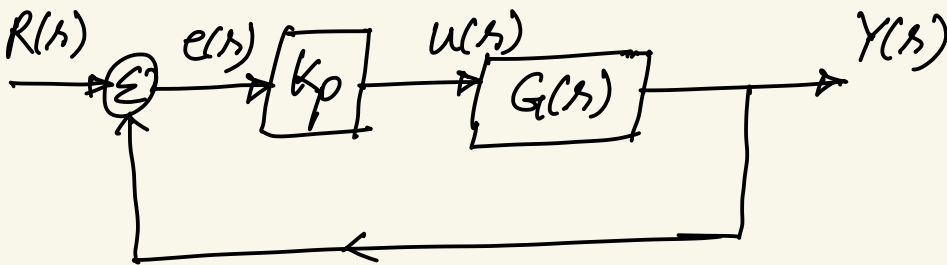
$$\frac{U(s)}{E(s)} = k_p + k_d s + \frac{k_I}{s}$$

The transfer function of a generic 2nd order plant is as follows

$$G(s) = \frac{b}{s^r + as + b}$$

We would like to see how the closed loop transfer function is influenced by the inclusion of the controller.

For the proportional controller, the block diagram is as follows



$$T_{cl}(s) = \frac{k_p G(s)}{1 + k_p G(s)}$$

$$= \frac{k_p \cdot \frac{b}{s^2 + as + b}}{1 + k_p \cdot \frac{b}{s^2 + as + b}}$$

$$T_{cl}(s) = \frac{b k_p}{s^2 + as + b(1+k_p)}$$

$$\left\{ \begin{array}{l} \omega_n = \sqrt{b(1+k_p)} \\ 2\zeta\omega_n = a \end{array} \right. \quad \text{dependent on each other.}$$

$$\zeta = \frac{a}{2\omega_n} = \frac{a}{2\sqrt{b(1+k_p)}}$$

ff k_p increases, then ω_n increases too. However, the ζ is reduced.

ω_n and ζ cannot be independently adjusted. They depend on

each other. Single degree of freedom system.

Effect of steady-state performance for proportional gain k_p :-

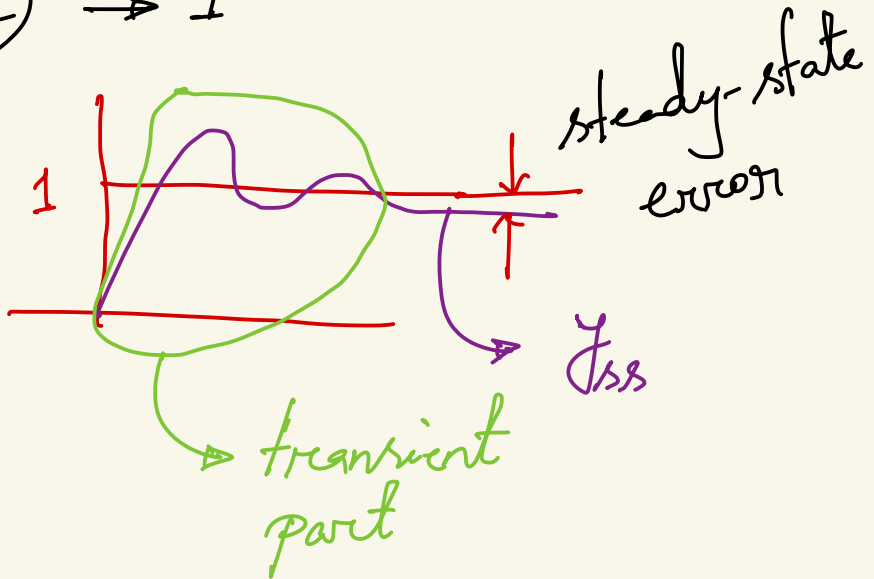
Steady-state value for a step input can be evaluated by employing the final value theorem (discussed in the class)

$$\lim_{t \rightarrow \infty} y_{ss} = \lim_{s \rightarrow 0} sY(s)$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{b k_p}{s^2 + a s + b(1+k_p)} \cdot \frac{1}{s}$$

$$= \frac{b k_p}{b(1+k_p)} = \frac{k_p}{1+k_p} \neq 1$$

As $k_p \rightarrow \infty$, y_{ss} (steady state value) $\rightarrow 1$



PD Control (proportional + derivative):

The closed-loop transfer function of the PD controller is as follows:

$$T_{cl}(s) = \frac{b(k_p + k_D s)}{s^2 + (a + b k_D)s + b(1 + k_p)}$$

This is not a canonical form but the trends are meaningful.

$$\omega_n = \sqrt{b(1+k_p)}$$

$$2\zeta\omega_n = a + bk_D$$

$$\Rightarrow \zeta = \frac{a + bk_D}{2b\sqrt{1+k_p}}$$

k_p and k_D can be independently adjusted to achieve proper damping.

Steady-state error of the PD controller:

$$y_{ss} = \lim_{s \rightarrow 0} s \cdot \left(\frac{b(k_p + k_D s)}{s^2 + (a + bk_D)s + b(1+k_p)} \right) \cdot \frac{1}{s}$$

$$y_{ss} = \frac{k_p}{1+k_p} \neq 1$$

As a result, with a PD controller steady-state error does not improve.

PD controller is a 2-degree of freedom system. We can place the poles of the system anywhere we like.

First adjust k_p values. Then adjust the k_d values to achieve the desired response.

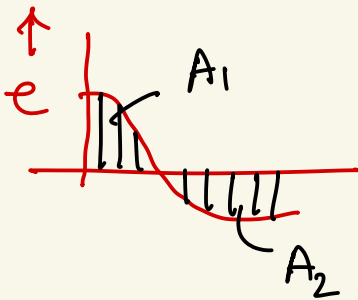
PI control:

$$D_c(s) = k_p + \frac{k_I}{s}$$

The closed loop transfer function of the PI controller is as follows

$$T_u(s) = \frac{b(k_p s + k_I)}{s^3 + a s^2 + b(1+k_p)s + b k_I}$$

This is a third order system.
It is usually challenging to make general comments.



$$\text{Sum} = A_1 + A_2$$

Disadvantage: control action does not change direction.

Windup situation may occur.

We may need to unwind the integrator
 for $\rightarrow$ subtract the error off
 $\rightarrow$ anti-windup

tend to be slower to react and the system may get oscillatory.
"saturation" may occur.

Steady-state performance of a PI controller :

$$y_{ss} = \lim_{s \rightarrow 0} s Y(s)$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{b(k_p s + k_I)}{s^3 + a s^2 + b(1+k_p)s + b k_I} \cdot \frac{1}{s}$$

$$= \frac{b k_I}{b k_I} = 1 \quad (\text{no steady-state error})$$

This is the biggest benefit of the PI controller.

A small K_I can make SS error go to zero. (longer time needed)